

WHY WE FEAR AI

ON THE INTERPRETATION OF NIGHTMARES

Hagen Blix and Ingeborg Glimmer

Why We Fear AI: On the Interpretation of Nightmares
© Hagen Blix and Ingeborg Glimmer

This edition © 2025 Common Notions

This work is licensed under the Creative Commons Attribution-NonCommercial 4.0 International. To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>

ISBN: 978-1-945335-17-4 | eBook ISBN: 978-1-945335-42-6
Library of Congress Number: 2024952077

10 9 8 7 6 5 4 3 2 1

Common Notions
c/o Interference Archive
314 7th St.
Brooklyn, NY 11215

Common Notions
c/o Making Worlds Bookstore
210 S. 45th St.
Philadelphia, PA 19104

www.commonnotions.org
info@commonnotions.org

Discounted bulk quantities of our books are available for organizing, educational, or fundraising purposes. Please contact Common Notions at the address above for more information.

Cover design by Josh MacPhee
Layout design and typesetting by Suba Murugan
Printed by union labor in Canada on acid-free paper

CONTENTS

INTRODUCTION	v
PART I	1
CHAPTER ONE CAPITAL, WILL, AND ARTIFICIAL INTELLIGENCE	3
CHAPTER TWO RESISTANCE IS FUTILE: THERE IS NO ALTERNATIVE	11
PART II	59
CHAPTER THREE WHO KNOWS THE FUTURE: AI, OPPRESSION, AND CONTROL	61
CHAPTER FOUR ON SUMMONING DEMON GODS: AI IN THE ECONOMY	79
CHAPTER FIVE WHICH SIDE ARE YOU ON: PROFESSIONALS AND TECHNOCRACY	121
CONCLUSION	167
ACKNOWLEDGMENTS	171
ENDNOTES	173
ABOUT THE AUTHORS	185
ABOUT COMMON NOTIONS	186

INTRODUCTION

“The reasons why we have got out of the habit of thinking of our history as progress are obvious. For even when twentieth-century progress is most undeniable, prediction suggests not a continued ascent, but the possibility, perhaps even the imminence, of some catastrophe: another and more lethal world war, an ecological disaster, a technology whose triumphs may make the world uninhabitable by the human species, or whatever current shape the nightmare may take. We have been taught by the experience of our century to live in the expectation of apocalypse.”

Eric Hobsbawm, *The Age of Empire* (1989)¹

A new form of magic is creeping into our world. It is the stirrings of a machine, of an inhuman silicon thing, beginning to act like it is anything but. Through the whirring metal casings that fill the newly overflowing data centers streams not just electricity, but language. For the first time, we can talk to something that is not a human—and that something talks back. Our questions flow into the machines in the data centers, and out flow words, sentences, paragraphs, and pages. At times, it almost feels like there is a someone in that box, not a something, a who and not a what. Whatever it is, a new kind of being is being born. Slowly, day by day, and word by word. Something is animating what was mere transistors before, something called artificial intelligence. A new kind of technological magic? Perhaps.

In science fiction, it is often said that any sufficiently advanced technology is indistinguishable from magic. In other traditions

(literary or otherwise), magic is associated not with technology, but with language, with special words or phrases. Stories are full of spell words, like *abracadabra* or *open sesame*, and when the right person uses them in the right context, they can have powerful effects. In myriad stories, from ancient myths to fantasy novels, simply naming things can bestow power, knowing someone's "true name" means you can control them, and speaking occult languages is associated with special magical talents. Even the ambiguity of the word "spell" betrays magic connotations: one can cast a spell, or one can spell a word (one can even spell trouble or doom).

And indeed, even for those of us who do not believe in magic, language surely has magical qualities. If you make just the right kind of sounds you can initiate a spooky action at a distance, a strange form of mind control. Make some random sounds, and you will attract, at best, strange looks. But if you know just the right "spell," just the right sounds—or just the right way of moving hands and body if your language is signed—and you know just the right recipe to combine them, you can conjure up new thoughts and images in the mind of another. You can transubstantiate the waves of sound or the electromagnetic waves of light into idea-waves. If you have learned to create the right shapes (say, by laying down some pigment on a surface, or chiseling away a stone), you can even talk to people in the future. Admittedly, even if you do master that particular kind of spell(ing), the people from the future cannot talk *back*. But, it certainly sounds like magic otherwise, doesn't it?

Artificial intelligence folds these two ideas of magic into one—it's the first kind, technology, that has begun to use the second kind, language. Up to this point, language was a human specialty, something often taken as a (or even *the*) thing that distinguished our species from all others. Now, it seems, these new machines are wielding their own incantations. What kind of strange thoughts do the machine magicians conjure up in our heads? For some, certainly, it is dreams of all kinds—from personal wealth to somehow solving climate change. But all too often, it seems to be worries, fears, forebodings, and nightmares.

Some of these worries are quite concrete: from misinformation campaigns to the environmental impacts of data centers, from

AI-encoded social biases to the effects of possible automation, AI seems poised to shape how existing social issues develop. In many cases, social conflicts about the path forward are already underway. Labor unions, for example, have been fighting against very concrete AI-driven threats to the livelihoods of their members and to the quality of their work. These worries affect a wide variety of workers, from script writers and actors to longshoremen: in 2023, SAG-AFTRA (the Screen Actors Guild - American Federation of Television and Radio Artists) and the Writers' Guild of America went on strike and won contracts that included restrictions on AI use, mandating that it cannot be used to replace writers,² and that using AI simulations of likeness, voice and performance must be consented to and fairly compensated.³ In 2024, the International Longshoremen's Association also went on strike, demanding limitations on the use of automation, a move that has been widely interpreted as part of a broader pushback against AI.⁴

If one looks around, however, the worries about AI do not stop at threats to ways of making a living. There is, in addition, the stuff of real nightmares, seeping into spaces that are not otherwise known for their flights of fancy. *Time Magazine*, for instance, published a bleak piece on AI development that warned "the most likely result [...] is that literally everyone on Earth will die." Illustrating the piece is a gif of a golden circuit diagram: against a flickering black and red background, it sprouts like a tree, circuits reaching out, first wide, then narrow then wide again, until the top blossoms into the shape of a mushroom.⁵ Through the flickering colors and the upward and outward growth, it evokes, beautifully and terrifyingly, a specter of a digital nuclear blast. The message is clear.

From humanities professors to billionaires, from AI researchers to spectacular fraudsters, from journalists to the pope, it can seem like all the kinds of people who commonly make it into our media discourses are experiencing equally nightmarish visions. Public statements and open letters warn that humans might be about to lose control or even go extinct. The philosopher of consciousness David Chalmers has said that "in the worst case, [humans are] extinguished entirely, [which] is terrifying."⁶ The pop-historian Yuval Harari has teamed up with others to

warn that AI could “hack the foundations of our civilization.”⁷ Perhaps the briefest of all the open letters is a single sentence: “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.”⁸ It was signed by, among others, the CEOs of three of the largest AI industry labs, a long list of academics and billionaires, and even former presidents and astronauts. And lest you think we’re just hand-picking doom-mongering crackpots on a totally arbitrary basis: at least two of its signatories have since received a Nobel Prize. Crackpots or not, the nightmares certainly are haunting people who are anything but peripheral to today’s centers of power.

Eric Schmidt, the former CEO of Google and Alphabet, is also evoking AI nightmares. He has turned into a wandering AI evangelist, a new role involving such stations as Chair of the US National Security Commission on Artificial Intelligence, as well as funding and/or sitting on the board of AI startups. Several of those are defense contractors, selling drones, target identification and/or surveillance systems (coincidentally, precisely the things that the government commission he chaired strongly advocated buying).⁹ Schmidt says AI is an “existential risk,” which in his words “is defined as many, many, many, many people harmed or killed.”¹⁰ Even the Pope, whose thoughts on artificial intelligence are mostly serious reflections on the effects that technological developments will likely have on global inequality, has also called AI “both a terrifying and fascinating tool” that “humans must not lose control of.”¹¹

AI nightmares may even be inspiring outlandish crime! A former American business executive, Caroline Ellison was involved in what prosecutors called “one of the biggest financial frauds in American history,” to the tune of \$8 billion. What grand plans motivated her to abscond with that much money? According to her former colleague, “Ellison’s drive to make money was the result of her [...] desire to use her wealth to stop artificial intelligence from causing humanity’s extinction.”¹² That sentiment was, apparently, shared by her more famous partner-in-crime, Sam Bankman-Fried, founder of the involved cryptocurrency exchange FTX. He, too, wasn’t driven by anything as base as greed,

when they stole billions of dollars from deposits. Bankman-Fried “needed infinity dollars because he planned to address the biggest existential risks to life on earth.”¹³ Among them: “artificial intelligence that turned on mankind.”¹⁴ Their surefire way of rescuing humanity from the pernicious AI menace was... building more AI. They invested at least \$500 million in ill-gotten gains in an AI company called Anthropic. Well, what’s a couple hundred million stolen dollars if you’re having nightmares about getting trapped in the Matrix, or facing the Terminator?

What is this new technology, this alleged “intelligence” that can talk, evoke such nightmares, and elicit such crass money grabs? At its heart, it’s a set of statistical techniques, called *machine learning*, that can be used to create mathematical abstractions over data—models. Machine learning algorithms do this by extracting probabilistic patterns from often vast amounts of data. The resulting models can be used to make predictions about new data. Machine learning can be applied to a variety of issues, say, predicting the three-dimensional shapes of proteins from their amino acid sequence, playing Go, identifying objects in images, or producing synthetic speech from text. Useful tools, one might think, and sometimes quite impressive ones—but hardly the stuff of nightmares.

What then gave machine learning methods like “deep learning” a connotation of dark and arcane arts, as if the word “deep” was a reference to raising Lovecraftian “deep ones,” and not just a word describing a property of artificial neural networks (perhaps the most prominent tool in the machine learning toolbox, they are called “deep” when their nodes are organized into more than two layers)? Or, to put it another way, why did the mundane statistics of machine learning become almost synonymous with the speculative and even grandiose term “artificial intelligence”? Perhaps one part is simply that applying machine learning techniques to language worked surprisingly well. There are now impressively good models of corpora, of huge amounts of text—language models. Machine learning has produced language models that can match probabilistic properties of the text in corpora, and use those patterns to create contextually plausible continuations of text. In short, machine learning techniques have enabled the

creation of models that can “talk” to us. And yet, this can’t be all of it—after all, language in itself is not scary. Someone being able to talk is not generally grounds for special suspicion or extinction nightmares. And computers gaining natural language interfaces in science fiction stories does not universally trigger fear either. Take, for instance, the bridge computer in Star Trek stories—it clearly understands commands of all kinds, from finding and cross-referencing information to setting variables of the environmental controls. And yet, neither viewers nor characters find such interfaces to be particular cause for concern. After all, language may be magic that can put ideas into our heads—but it’s not mind control, we aren’t forced to believe those ideas.

Perhaps a straightforward, cynical take is the right way to go: maybe both the stoking of fears and branding of statistical tools as “AI” are simply part of that special hyperbole called marketing. After all, if what you’re selling could pose an existential risk, it must be very powerful indeed, and maybe its powers could be harnessed for good as well. It calls to mind J.R.R. Tolkien’s *The Lord of the Rings*, a story paradoxically beloved by Silicon Valley. Paradoxically, because the ring in that story is indeed powerful—and it must never be used, because its power corrupts. Alas, Silicon Valley and its AI peddlers aren’t about to let such “subtleties” dampen the mood—in fact, they love naming their companies for things and characters from Tolkien’s Middle Earth. There’s Peter Thiel’s defense and surveillance company, Palantir (named after a magic stone that allows the user to see other places—though what it shows can be deceptively selective). There’s Palmer Luckey’s AI, robotics, and defense company, Anduril Industries (named after a sword). Even our wandering AI evangelist Eric Schmidt has funded a Tolkien themed company: Istari (the elvish word for wizards) is another startup that aims to use machine learning for weapons development.¹⁵ A sales pitch that sounds like “look at this powerful thing I’m about to build, it might destroy humanity!” may seem strange—but if you’re peddling military equipment, destruction of humans on a large scale is exactly what the customer is looking for.

It’s not just AI companies building weapons that deploy such scare-sales tactics though. Sam Altman, the CEO of OpenAI (the

company behind the chatbot ChatGPT) is well-known for trafficking in stark dichotomies. Talking at a 2023 venture capital conference (and hence presumably marketing to possible investors), he described what he saw as the possible future of AI: “I think the best case is like so unbelievably good that it’s hard for me to even imagine [...] just so unbelievably good that you sound like a really crazy person to start talking about it and the bad case and I think this is like important to say is like lights out for all of us.”¹⁶ Why does OpenAI want its investors to believe in unimaginably dangerous AI powers that could either kill all of us or, maybe, be harnessed for good (or at least for a tidy profit)? Well, they are hoping to build and sell, if not the One Ring, then certainly the One Tool. OpenAI is among a handful of companies that intend to sell more than simply some bespoke statistical models, or a better interface for computers that uses text and speech instead of mice and keyboards. No, the ultimate aim—so we’re told—is to produce not just artificial intelligence, but artificial *general* intelligence (AGI). What’s that? Well, any tool can be said to “surpass” humans in some domain—whether we’re talking about a hammer, a microscope, or a calculator, we invented those tools so we could do something that we couldn’t do before (or couldn’t do as fast, well, painlessly, profitably, etc.). And machine learning-based tools for particular tasks are no exception. But what if there were a tool that’s better than people at everything? That’s what AGI is supposed to be. In the words of the OpenAI charter—“highly autonomous systems that outperform humans at most economically valuable work.”¹⁷ Or in other words—one tool to rule them all.

What better way to make AGI seem imminently achievable than to say the machines are really scary—so scary that they might end up destroying us all (unless the Great Men™ who are building them retain control over them and manage to steer them in the right direction). Certainly, as a marketing and PR move, the scary AGI pitch has a second advantage: if the stakes are as high as the very survival of *humanity*, then concerns about the effects that really existing artificial intelligence technology has right now—from excessive energy usage, to undermining the livelihood of artists and dock workers, to introducing cybersecurity

vulnerabilities, to perpetuating and worsening discrimination, to disinformation and deep fakes, to supercharging mass surveillance, etc., are easily brushed aside.

So, there's certainly good reason for skepticism when we encounter AI fears and nightmares in the media. And yet, even for those of us who think that the hype about AI's future abilities and the scaremongering about its dangers are part of a sales pitch, questions remain. Why do media outlets publish such stories as "The End of Humanity: Will Artificial Intelligence Free Us, Enslave Us—or Exterminate Us" (*The Times*), "AI could go 'Terminator'" (*Fox News*), "AI could pose 'extinction-level' threat to humans and the US must intervene" (*CNN*), or "A.I. May Save Us or May Construct Viruses to Kill Us" (*The New York Times*)?¹⁸ Perhaps more importantly: what makes such stories resonate—especially with those of us who own neither crypto-exchanges to defraud people with, nor chatbots or automated weaponry that they want to sell? What do the nightmares of the world's Eric Schmidts and crypto-fraudsters have to do with the worries of the union workers?

Maybe the OpenAI definition of AGI can provide us with a crucial hint—particularly the "economically valuable work" part. The science fiction author Ted Chiang remarked in an interview that, perhaps, "most fears about A.I. are best understood as fears about capitalism [...] as fears or anxiety about how capitalism will use technology against us."¹⁹ If that is true, then the nightmares of billionaires like Eric Schmidt and those of the union workers must be about rather different things. After all, Eric Schmidt is far more likely to find himself on the side of those who do the using, rather than on the side of those that the technology might be used against. That is, of course, the nature of capitalism: some people own vast amounts of assets (e.g., shares in companies, real estate, patents, and, of course, technology), and they use those assets to make more money, and more assets. The rest of us, those who do not own assets that can make more assets (or *capital*), make a living by working for those people and institutions that do. Whether it's a forklift, or a computer cluster—capitalists use their ownership and control over assets, over technology, to direct the work that the rest of us do for a wage, so that they can make a profit. In

other words, capitalism is a system with social classes, and what class someone belongs to changes how they relate to technology.

If we want to understand AI nightmares, then, we must interpret them with class in mind. Perhaps even the very same myths about AI doom may turn out to ring true to different people for quite different reasons—we'll each have to interpret our own nightmares. We'll have to figure out who gets to be considered part of that vague and elusive "we," of "humanity," that future AI supposedly threatens, and who is merely an afterthought. We'll have to make sure not to get taken in by the fables of the crypto-fraudsters or the weapons peddlers (some of whom already weave the very nightmares that we live in). Let's get into the thick of the politics and economics of AI—and interpret AI nightmares against that backdrop. Along the way, we'll encounter an array of colorful characters from autonomous technology with its own will, to steam demons, from technological deities to Friedrich Hayek's ghost. We'll see how capitalism and class shape both the actual tools called AI and the AI nightmares. And of course, we'll try to find, among the nightmares, inspiration for a better future.

CHAPTER 4

ON SUMMONING DEMON GODS: AI IN THE ECONOMY

Just as states make use of AI technologies, so do companies. Perhaps unsurprisingly, they sometimes use the very same technologies. The US pharmacy chain Rite Aid, for instance, used facial recognition software from 2012 to 2020, in order to identify people and classify them as potential shoplifters. Just as the policing algorithms misidentified people, Rite Aid's algorithm "falsely tagged consumers, particularly women and people of color, as shoplifters," according to the Federal Trade Commission.¹ Obviously, privatized security will be interested in many of the same tools as the police. And certainly, companies will be interested in tracking their customers with AI, whether as targets of ads or as imagined thieves. For most companies, however, there is an even bigger target at which to point their AI technologies: the people they employ. When companies do so, the ostensible purpose is usually simply described as increasing productivity. After all, who could be opposed to getting more done with less work? Alas, increases in productivity are usually deeply interwoven with two other purposes: first, the automation of supervision and control—management. Second, the reduction of wages, for instance by increasing the pool of workers that can be hired for particular tasks—deskilling, outsourcing, and globalization. These are, in other words, uses of AI that are intimately linked with class. Let's explore those uses, those links—and some nightmares they give rise to.

We'll get into the weeds below, but we can already preface something about the kinds of nightmares relevant to the experience of AI as automated management, and AI as a deskilling tool. So far, we have mostly talked about nightmares of the destruction of humanity. There is, of course, a second class of common nightmares—perhaps more Matrix than Skynet and its Terminator. In these nightmares, AI comes to dominate and control our lives, rather than trying to exterminate us. Take, for instance, a series of comments that Nick Bilton of *Vanity Fair* reported in 2023: one AI engineer claimed “We’re creating God.” Another AI insider is quoted saying “With AI, we’re summoning the demon.” A fellow journalist sums up a widely held perspective as follows: “They believe the AI will remake itself into a god, with godlike powers, and in a dystopian act of supremacy, the gods will overwhelm us and take our place. So now we have to appease the AI gods.”²

How should we think about these fears of domination by AI demon gods? At the moment one starts to think about not just humanity, but its actual divisions, domination becomes relevant. Domination of people by people, that is. And when we think about the division called class, it is clear that a relation of domination is involved. That is, after all, part of what class is all about: hiring a person is quite unlike getting a job. No matter how much self-fulfillment, self-discovery, or community the PR and HR departments of some companies may be promising, to hire oneself out is not the same as engaging in a shared project with other people on an equal footing. Clearly, the relation between a capital and a worker, between a boss and an employee—whether there is fluff around it or whether it is nakedly obvious—is a relation of domination, of controlling another’s life, of one person telling another what to do. Let’s jump into AI, class, management, supervision, deskilling, and domination.

THE STARRY EYE ✨ ABOVE US: AUTOMATING OVERSIGHT

Let’s start from supervision, and let’s take a rather large step back, to think about management in the broadest possible terms: most

things that we call power, rest ultimately in the ability of some person or group to systematically extract the fruits of labor from another group. From conquest and plunder to forced labor, from feudal dues and taxation to waged labor, across history, such power has taken many forms, in many different systems of classes. We might say that, throughout all these systems, one can distinguish two very different ways of obtaining those fruits of labor, one which requires something akin to management, and one in which the extracting class can take hold of the fruits of labor directly, whatever they may be. Think of a feudal lord who looks around, and simply demands a certain amount of eggs, wheat, barley, etc. from his peasants (perhaps as a fixed amount, or as a certain percentage of what is produced). Or think of the early capitalist putting-out system, where a shoemaker or a tailor might be hired and supplied with resources, to produce clothes, or shoes, in their own workshop. They were paid per completed piece (pair of shoes), and the capitalist who hired them took hold of their fruits of labor (surely representing a higher value to the capitalist, than its costs had been—where else would profit come from). Taxation has often taken a similar structure, and if we stretch our categories a little, we might even think of a modern income tax as an extension of these older systems, though here the state takes not a slice of the fruits of labor, but rather a cut of the worker's pay. The point is—one can engage in obtaining the fruits of other people's labor after the fact, without being at all involved in the production of those fruits. In other words: without any management.

In the other kind of situation, however, the way the powerful benefit from others' labor looks quite different: rather than taking control of the fruits of labor after the fact, the extracting person, class, or institution may take control not just of the things produced, but also of the person producing, of their ability to work, and direct that ability towards the extractor's own goals. For example, when the ancient pharaohs wanted to have some giant pyramids, they looked around and couldn't find any. To their misfortune, they realized that they did not have the same option as the feudal lords would have thousands of years later. The pharaohs couldn't simply take some pyramids that people had

built of their own accord, in the way that our feudal lord could later grab some grain. They had to make people build pyramids. They had to take control not simply of the fruits of other people's labor, but of the process of labor itself. One of the peculiarities of this kind of situation is that it generally also makes a new kind of work necessary: the work of supervision. What is a supervisor, if not someone who represents the interests of those for whom the work is done within the labor process, in opposition to those who actually do the work (whether it concerns assembling pyramids or computers). Throughout history, supervisory work must have had more names than we know—foreman, overseer, superintendent, academic advisor, or manager are just a few (some more, some less euphemistic).

Under capitalism, those who own assets for the purpose of generating more assets, are rarely content with merely snatching the fruits of others' labor. Capitalists and capital are deeply involved in the production of objects and services alike. Bosses give orders and directives. The machines in the factories or coffee shops, the cubicles and office hallways, and the computers and algorithms that we use during work, all direct us in their ways. And wherever these things belong to someone other than those who work with them, they are capital. Sandwiched in between all this capital, we live in a world not just of ubiquitous supervision, but of recursive oversight, a world of managers of managers of managers. Capitalism has constructed veritable pyramids of managers, trying to solve a recursive problem: in controlling the process of work itself, capital finds it always needs to control the individual workers, who—to its eternal dismay—continue to be human beings with desires of their own, not all of which align with the "will" of capital, the drive for profit. These humans with their pesky needs do horrid and evil things, like taking breaks. They seek comfort, and perhaps even dignity.³ Hence, they need to be watched, lest they get away with it. And what does poor capital have at its disposal to watch people? Why, more people! Now hired as watchers—but alas, who, the frantic capitalist immediately wonders, watches the watchers? So, watcher-watchers are hired, and watchers of those. As capital grows, so grows the metaphorical pyramid of surveillance, oversight, and management.

As capital goes about making the whole world its comfortable nest, those pyramids, however, are a thorn in its side. After all, companies pay people to make things that can be sold, but the watchers just watch, instead of making. They are simultaneously necessary, and superfluous—and nobody likes a paradox. Throughout its short history, capitalism has, again and again, shown a special adeptness at a strange transformation, which will help it get out of this mess: it transforms tools themselves into things that can enact a degree of supervisory control. Tools can now substitute for watchers. We will get to the use of artificial intelligence in exactly this kind of context, as an automated managerial oversight, in just a second. But it turns out that looking first at an earlier transition of the same kind is strikingly illuminating about many properties of the current historic moment. That transition, with its then-brand-spanking-new technology created a surprisingly similar media flurry. When they were describing that technology, people spoke of “an intelligent being, devoted to our services,” and even “a being [...] of supernatural might” that “stands as if it had life and breath in it.”⁴ One would be forgiven for thinking these quotes were about AI. But they are from the early 1800s and describe—the steam engine.

We can see a curious similarity, between then and now, between media discourses around steam and those around AI. Media discussions of AI, it seems, are more often than not structured around a duality of fears and hopes. Wherever one looks, it’s “promises and perils,” “will artificial intelligence free us, enslave us, or exterminate us,” it’s about “it serving us, or us serving it,” and AI futures being “unimaginably good, or lights out for all of us.” When it comes to steam, we find basically the same phenomenon—a fact that should come as a surprise to those who think AI fears are about an alien intelligence, rather than capitalism.

Back then, Frederick Marryat, a Navy officer, wrote about steam, “I never can divest myself of the idea that it is possessed of *vitality* [...] will this powerful invention prove to mankind a *blessing* or a *curse*?”—his emphasis! In his book *Fossil Capital*, the historian Andreas Malm points out that the reactions to steam are often quite class-aligned. The capitalists, and those paid to extoll their virtues, praised the steam engine as a “tool [that] did

all the working (for the thinking had before been embodied in it),” and suggested that it had “within itself, almost a creative power.” They opined, in quasi-religious tones, of “the mighty Spirit of Steam,” and claimed that “of a sudden, a soul had been put into that wonderful creature of man’s making.” Steam even “summons around him [!] his myriads of willing menials.” If it weren’t for the occasional archaism in the syntax, it would hardly be surprising to read any of these things being written about today’s AI technologies. What did the working classes write? In their newspapers, they called it “a tyrant power and a curse to those who work in conjunction with it,” and “our enemy [...], the servant of the rich man.” They wrote poetry that called steam a “demon god of factory and loom,” or a “ruthless king.” In one poem, the machine speaks: “Foe, fau, fum—// I smell the blood of the working man; // Be he alive, or be half dead, // I’ll grind his bones to make my bread.”

The introduction of the steam engine came with its own version of “promises and perils,” its own tales of autonomous machines. Vague ideas about “intelligence,” as they are so widely entertained or spun in the media discourse, hardly provide an explanation for the parallelism (unless, perhaps, one really does believe that both AI and steam engines have a soul, a spirit, are thinking, and have creative power). But if it is the effects that these machines, be they steam engines, or AI models, have on class relations, on the way capital can exercise its power—perhaps, we can make sense of the allure of myths of “demon gods” (then) and “the true AI demon”⁵ (now), with their “supernatural might” (then) and “godlike power”⁶ (now). In both cases, a direct relationship between humans, between a supervisor/manager and a worker, has been replaced by a relationship mediated by a machine. And in both cases, the machine embeds capital’s need for control, the very same need that gave rise to the need for management to begin with. The worker experiences this control as a property of the machine, experiences the machine as dominating them. What else, but this story of technologically mediated domination—of humans and by humans—could the demon gods of steam and transistor be telling?

Insofar as the things people said at the time resemble what we find in the media now, we should prick our ears up and dig deeper—the similarity might be more about something the two situations have in common than about the two technologies in particular. The similarity might reveal something about the nature of technological change under capitalism, its driving forces, its effects, and even its relation to such things as control, power, knowledge, and agency. We'll explore how supervision and management can get embedded into tools, and compare AI to the steam engine to better understand capitalist economic constraints on technology—the way the will to profit manifests itself in the steam engine and AI alike. Let's dive into the past, so that we cast the lights of the steam engine's coal fires onto the development of AI today.

MACHINES CONTROLLING PEOPLE

Until 1784, wherever we look, the spinning of fibers into yarn was powered by one and the same energy source—people. Whether the fiber was wool or cotton, whether it used time-honored and traditional tools or the new crank-powered Spinning Jenny, the machines were powered by the movement of their operators. This was true where the spinning was for private use, and where the spinning was for a tribute or tax that had to be paid. It was true, even where production was already capitalist, as it was in Britain's putting-out system, where a merchant supplied the raw resources (especially cotton, which had to be imported), and hired spinners to perform the work at home. In that year 1784, however, a British merchant named Samuel Greg opened a new kind of cotton mill near Manchester, one whose yarn-spinning machines were powered not by human hands, but by water power. As Sven Beckert puts it in *Empire of Cotton*, the hills around Manchester were the Silicon Valley of their time.⁷

This new source of power certainly increased productivity, making it possible for a worker to spin more yarn in the same amount of time than they could using hand-powered spinning tools. And yet, water power would almost immediately be replaced by another power source, steam. In *Fossil Capital*, Malm gives a

fascinating account of this transition. While coal had been burned for a variety of purposes for ages, the modern fossil fuel economy begins in the early nineteenth century with the widespread adoption of the coal-powered steam engine in the expanding British cotton industry. In common conventional tellings of the adoption of steam and coal, instead of water power is purportedly driven by two factors: first, only steam could provide the additional horsepower that the early industrial revolution required. Second, the steam engine supposedly provided an exceptionally cheap source of power. But Malm argues, drawing on contemporary sources, that looking closely at steam's main alternative—water, which powered that mill built by Samuel Greg—proves otherwise. As it turns out, water power could have easily been expanded, perhaps by as much as a factor of 20. So it cannot have been that only steam could have provided the extra horsepower required for expanding production. Even more surprisingly, water power was actually significantly cheaper than steam power at that time. So it cannot have been a price advantage either. What were the real reasons for the adoption of steam power then? According to Malm's research, steam had two major advantages for capital, and both were related to capital exerting power over its workers. And both advantages, as we will see below, also hold for today's AI technologies.

The first advantage of steam for capital was rooted in its spatial flexibility: while water mills obviously require a nearby river, steam engines can power a workshop anywhere one can bring coal. Most workers lived in cities, and steam engines could be built there to enjoy a large and easily accessible labor pool. And if workers went on strike in a city, they could much more easily be thrown out, and replenished from the larger supply of urban residents. The rivers needed for water mills, however, were often in places with comparatively few residents, meaning mill owners frequently struggled to find workers, and had to entice people to move from the cities. Some of them even constructed whole company towns, in order to get people to move closer to the mills. To the watermill owning capitalist's dismay, this situation provided the workers in a particular mill with collective bargaining power: because replacing workers was a costly affair,

watermill owners couldn't use the methods that their steam mill owning counterparts could: breaking strikes by simply throwing out a part (let alone all) of a mill's workforce. As we all know, in a world driven by supply and demand, a worker who's harder to replace (i.e., lower supply) can demand a higher wage, or even shorter working hours. Steam may have been more expensive than water power when reckoned on its own, but steam also reduced the cost of labor, by increasing its replaceability. It made it easier to fire one worker and hire another, it enlarged the pool of potential workers for a particular job, and it increased labor replaceability. In other words, the steam engine was not just a tool and energy source for workshops, but also a tool for depressing wages. We'll explore below how wage depression in numerous forms is also one of the primary things that makes AI attractive to capital today.

In addition to its spatial advantage, steam also had a temporal advantage over water: with water-powered mills, speed was always to some degree dependent on the flow of the river, with the mill running faster when the flow was strong. But steam engines could be sped up almost at will. This became decisive when the British state limited the length of the working day. Amid waves of worker unrest and riots triggered by the new factory systems (whether driven by water or steam), the government intervened, and enforced a kind of class compromise between the interests of the capitalists and those of the workers. Fearing that these riots—forms of collective bargaining, as the historian Eric Hobsbawm points out⁸—would turn into uprisings, the government decided it had to reign in the factory owners to calm the rioters: it passed laws to restrict the maximum length of the workday, first to twelve hours, then to ten and a half hours a day. Before this point, owners of water mills had made up for the occasional lull in river flow rates on one day by adding work hours on a later day with a fast flow rate, demanding workdays of up to eighteen hours in length. But now the law limited the capitalists' license to draw out the maximum amount of labor from workers by increasing the length of the workday. The steam engine, but not water power, could make up for what was to the capitalists a terrible and unjust curtailment of their freedom: when the working day was shortened by, say, 20 percent, the owner of a steam engine would respond

simply by making the work more intense, stoking their machines to run 25 percent faster, and exhausting the workers just as much as before (or more) but in fewer hours, thereby stabilizing production and profits—or even increasing them.

When we look at this from the perspective of control over the labor process, from the perspective of supervision, we see the completion of the peculiar transformation that we ascribed to capitalism a few paragraphs ago: at first, the workers themselves set the speeds, powering the machines with their bodies (say by turning a crank to operate a Spinning Jenny). As capitalism became more widespread, a necessary conflict of interests arose between capitalists and workers—with capital seeking the maximum speed in its insatiable quest for more output and more profit, and with workers seeking a safe, or even (gasp!) comfortable work speed, caring perhaps more for self-preservation (and fingers) than for the profit of others. The conflict thus necessitated something else—supervision. Someone, either the capitalist himself or someone hired for the purpose, had to continually watch the workers, and push them to go faster.

But the water wheel turned this managerial problem into a technical one. It took the power to control speed away from the worker, the speed instead being set by the river and the machine. The steam engine completed this strategic move: it handed the power of speed-setting fully to the capitalist. Unlike the water wheel which took its speed from the river, the speed at which the steam engine ran could easily be set by the capitalist. The paradoxical status of needing a person to supervise the speed of the workers was resolved: as steam took over—what was once oversight by a human supervisor, now looks like a property of the machine. And even though these machines were, unlike today's language models, not "talking," we observe a familiar pattern, already existing in the past: with both yesterday's cotton mills and today's AI, the power of the owners can easily disappear from sight, appearing instead as the power of a machine, the power to compel people to work at a certain speed (that is, faster).

Why did the machine "possess" this power over people, this power to compel them? Obviously, it is not simply the machine itself—if the workers themselves owned the machine, they could

set the speed, and there would be no sense that the machine had power over them, any more than any other tool does.* But, alas, it was the capitalists who owned the machines, and set the speed. In fact, they introduced the machine in the first place, because it afforded them the power to set the speed. As such, the power accrued to them precisely as owners of such tools, as distinct from their users. These new machines were both (now) necessary in order to produce commodities at marketable prices, and at the same time were too expensive for workers to acquire. Hence, the workers had to find employers, in order to live. That is, of course, the modern condition of life: today, most people can only make a living, by hiring themselves out to someone else who is wealthy enough to own capital. Thus, power constantly seems like the property of capital, and where a capital consists of machines, machines can appear to wield power. It's as simple as that. And as strange.

MANAGEMENT WITHOUT MANAGERS

The steam engine allowed the capitalists of yesteryear to set the speed of the machine to their liking; today, AI systems are used for similar purposes. AI systems that process image data can be used not just to identify faces (as in policing), but also to classify movement. Workers in Amazon's warehouses, for instance, are constantly surveilled, measured, analyzed, and classified by an AI movement tracking system. At any step, stowers who shelve packages are monitored by cameras connected to AI technology

* Of course, even if they owned it in a cooperative way, but in a generally capitalist society, the workers would still have to compete with capitalists in the same market, and hence they would have to set the speed in a way that allowed them to produce competitively priced commodities for the market. In such a context, workers would be the ones setting the speed (again, that is, faster), and they would know it. But it would simultaneously be true to say that the market was setting the speed, that it is not simply the relation workers have to an individual capitalist, but to the whole structure of production for markets and profits, that pushes for ever higher speed. Only if there were no competition from profit-driven enterprises could the workers really be said to be collectively in control of the speed.

that measures speed and accuracy, registers item locations, and logs any shelving errors.⁹ These AIs allow Amazon to control the speed of work, as the steam engine did. They enable the semi-automated tracking of people's movement, down to the minutest details, so that management can squeeze the maximum amount of work out of them.

As the steam engine handed the power to set the speed to the capitalist, it reversed something in the relationship between tool and tool-user. The speed being fixed independently, the tool was no longer (experienced as) simply an extension of the workers' powers and abilities, but as a thing that controlled them. Amazon's AI system acts akin to the steam engine. Its ultimate goal is to direct the movement and speed of the workers. It flags violations of "stow etiquette" (the way workers have to shelve packages), and logs bathroom breaks. It times whatever the workers do, generating "stowing metrics" that tell management how long each worker takes per package. Workers who are "too slow" are threatened with warnings, write-ups, and ultimately, with being fired and replaced. These threats often come from another semi-automated system behind which the human management can hide, from behind which the representatives of capital can see without being seen.

Brian Merchant's book *Blood in the Machine* tells the story of Luddite resistance in the 1810s, when they were fighting against the unilateral introduction of machinery that destroyed people's livelihood in the name of profit. He also describes parallelisms between the introduction of new technologies then and now. On semi-automated human resources algorithms, he quotes Veena Dubal, a law professor working on platform-based labor like Uber or Amazon: "Algorithmic hiring and firing have become an everyday part of people's lives, whether we're talking about customer service agents or warehouse workers." In her estimation "the aim is to create an anxious, uncertain workforce that has no choice but to be malleable before the algorithm's demands."¹⁰

As Amazon automates the surveillance of its workers, it replaces the bottommost level of its management pyramid with AI, with cameras and algorithms. But in the warehouses, according to Amazon's own estimation, its supervisory algorithm is only sufficiently "certain" of its classification about 95 percent of the

time. The remaining five percent represent thousands upon thousands of videos, anywhere from two seconds to two minutes in length. And they remain in that vestigial state of the managerial paradox—needing human watchers. They are sent to workers in places where labor is cheaper, to India or Costa Rica—mirroring once again the “spatial advantage” of the steam engine, which could be placed near the largest and most desperate labor pools of yesteryear.

Amazon’s AI system depends on splitting labor into these two halves—movement and the tracking of movement. This split labor is globalized, as in split literally across the globe. AI divvying up myriad tasks, distributing them among workers employed in supply chains wrapped around the world, is not just a consequence of AI technology newly coming into being, it is also AI replaying a familiar property of global capitalism. The adoption of the steam engine and the industrial processing of cotton in Britain was obviously predicated too on an earlier version of capitalism’s global supply chains, relying on colonization and its accompanying violence to supply the very cotton that was processed in Britain’s steam-powered cotton mills. Indeed, Samuel Greg, the manufacturer who installed that first water wheel in 1784, was deeply embedded in the networks of what Sven Beckert calls *war capitalism*, i.e., forced labor, combined with the state’s ability to violently seize land, and force the opening and creation of markets abroad. Samuel Greg owned not only that water wheel in Britain, but also plantations and enslaved people in the Caribbean. His immediate relatives were involved in various colonial enterprises, as cotton traders, plantation owners, and slave traders. When it comes to Amazon’s AI algorithm, we find that the networks of those earlier phases of colonial capitalism, though vastly reconfigured, never went away. Today, the old-new channels of empire, which now take the form of underwater cables and satellites, make available the massive amounts of cheap labor that so many AI systems depend on. Clearly, the old networks of colonialism are put to ever new uses that serve the needs of today’s capital.

Today, it is supervision itself that is produced by an industrial and partly outsourced production process: low-waged workers abroad watch thousands of videos every day (during peak times

in eleven hour shifts), classifying the activities of stowers in the warehouses. Like the warehouse workers, those in the surveillance rooms are also timed, each second measured, each tiniest task mapped out—and in both places, workers are penalized for any deviation from what Amazon deems optimal. In the warehouses, people cannot make a false move without getting flagged—in the surveillance centers, people cannot blink without missing targets, according to their own descriptions: videos have to be analyzed and categorized so fast that the literal blink of an eye can negatively affect their target accuracy. Since neither side is engaged in any activity that would plausibly make them count as managers, we might call this setup the industrial production of management without managers.

Such an industrial production of oversight is, in many ways, the result of the recursive application of managerial theory to management itself, of treating the production of oversight in just the same way as capitalism treats the production of everything else. What do these managerial theories look like? Around the turn of the twentieth century, a school of managerial thought came to prominence which has influenced many subsequent managerial schools from Fordism to today's Just-in-Time Production, and all the other ones that are taught to people getting MBAs or the like. That school is Taylorism, or scientific management, named for its (in)famous twentieth century proponent Frederick Winslow Taylor.

In his 1974 book, *Labor and Monopoly Capital*, the sociologist Harry Braverman describes the principles of Taylorism as the “explicit verbalization” of capitalism, and more narrowly of the interests of capital.¹¹ Taylorism is an attempt to make explicit and codify the managerial practices inherent to capitalism, and to turn management into a scientific approach to labor control and labor division. Like his precursors (writers like Adam Smith or Charles Babbage), Taylor wanted to make explicit and optimize what was implicit in the actual practices of earlier capitalists—practices that we have seen in the adoption of the steam engine.

The first and central principle of Taylorism, according to Braverman, is concerned with consolidating and centralizing knowledge of labor processes within managerial structures. As

Taylor himself puts it in the section of *The Principles of Scientific Management*: “The managers assume [...] the burden of gathering together all of the traditional knowledge which in the past has been possessed by the workmen and then of classifying, tabulating, and reducing this knowledge to rules, laws, and formulae.”¹² As capital involves itself in production—through its principal agents, like management—it aims to take over all knowledge of the labor process: who does what, how do they do it, with which tools and raw materials, what are the steps, when do they do it, how long does it take—and how much does everything cost. And it aims to use that knowledge to intervene in and control the labor process.

To figure out the best ways to intervene in work and gain advantage, to learn what the maximum speed is at which any particular task can be performed (to establish it as a standard, forcing everyone to work faster—perhaps even using a steam engine), Taylorist managers conduct, for instance, time studies to gather knowledge of how long each step of a labor process takes, or motion studies to investigate and classify the particular bodily motions involved. In Taylor’s time, they used tools like stopwatches or stroboscopic pictures to transfer knowledge of labor processes from the workers (who did, after all, already know how to perform their work) to management, as per Taylorism’s first principle. In the Amazon AI, this Taylorist drive has itself become the object of an industrial production process, where the chain of AI and video watchers continually produces the modern equivalent of time studies and movement studies, of automated oversight.

If attaining control of the labor process is the means, capital’s goal is to minimize the costs per unit produced (whatever that may be—perhaps it is even oversight itself). Or in other words, the goal is to minimize wages (lowering wages per unit time), and to maximize productivity (that is, lowering the amount of time it takes to produce a unit of commodity). It’s one of the more peculiar things about capitalism that these two things—which might be judged quite differently with respect to their ethical implications, or their social desirability—look quite the same to the capitalists, as both are simply ways of reducing the cost per unit. These two goals are, of course, not inventions of Taylorism, but inherent to capitalism—lower costs mean higher profits, after

all. Indeed, the steam engine was adopted precisely because it allowed the capitalists to pursue these very same aims: wages were minimized by making it easier to hire and fire workers, by making workers more easily replaceable; productivity was maximized by giving sole control over speed to the capitalist, by speeding up work and reducing the unit time needed to produce a unit of the commodity.

Taylorism, of course, is a theory of management, not of technology. While the steam engine was a technological way of achieving the goals of capital (and thus a way of embedding capitalist class relations right into our tools), Taylorism is a managerial approach to the same goals. This managerial approach, just like technological approaches like the steam engine, seeks to make workers more replaceable in order to lower wages. But while the steam engine's ability to do so came from the fact that it could be placed near large groups of people in need of waged work, the organizational/managerial method is to break down complex tasks into many smaller steps. In Amazon's warehouse AI, we see both methods combined. In Taylorist fashion, it separates out the classification of individual stowing actions from other managerial tasks. And like the steam engine, it then takes advantage of new spatial possibilities, distributing the labor of watching videos around the world, to wherever the lowest wages can be found.

In separating out a part of the managerial tasks, the Amazon warehouse AI is used to separate so-called skilled labor from so-called unskilled labor—that is, tasks that require less/cheaper training, or for which there is simply a larger pool of potential hires, can and thus will be paid less. This is often called *deskilling*. Regardless of what those who purchase people's time and skills may say, it is of course not called "deskilling" because the subdivided tasks do not require genuine skills, but because they do not have to be paid as well as so-called skilled labor. It's all about increasing the supply (and thus decreasing the cost) of the particular skills and abilities that companies need to buy on the market. In this case, what is being deskilled is actually part of management itself, which is transformed into an industrial production process of video classification. This result of applying managerial theory to management itself is the techno-Taylorist

production of Taylorist management and knowledge gathering, the automation of its own prescriptions.

INDUSTRIAL PRODUCTION OF OVERSIGHT

Amazon's industrial oversight production may even be auto-surveilling: the video watchers, too, are on camera. Niamh McIntyre and Rosie Bradbury of the Bureau of Investigative Journalism quote Mateo, a former Amazon video reviewer, who said he would sometimes see videos of rooms just like the ones in his building: "Probably someone else, somewhere else, was watching me at the moment I was watching them." Capitalism's problem, "who watches the watcher," previously led to an expanding pyramid of management. This worked in practice, but it continued to create the same paradox, insofar as every solution to the problem produced merely a new set of watchers, and hence posed the same problem again. This is, of course, still true today. After all, the metrics generated by Amazon's system are still put in front of managers, it is only the bottom-most layer, the one closest to the workers themselves, the actual immediate watching, that has, so far, been transformed from a managerial relation into a semi-automated production process. Nonetheless, it is worth asking whether the auto-surveilling nature of that bottom layer is perhaps a truly new attempt at solving the paradox, and how far it might spread up the management pyramid. No matter where and how far it will spread, however, we can already assume with little doubt that the PR machines will call this dystopian attempt at a new solution all kinds of lovely names, declaring the economics of it circular, its decision making decentralized, and its hierarchies flat.

From the perspective of Amazon, this industrial production of oversight by the video watchers is also doubly useful. As its immediate purpose, the video classification serves directly to discipline and control the warehouse workers, whose behavior is labeled. At the same time, the videos and the labels constitute the very data used to make and refine the surveillance AI itself. We can make some educated guesses here about the way the Amazon warehouse surveillance AI is trained. In all likelihood, Amazon is using a common procedure to make the maximum use of the data

that it has collected from warehouse workers and video watchers. First, the raw video data of warehouse stowers can be processed using a training paradigm known as *self-supervised* or *unsupervised* learning. At this stage, the machine learning system extracts probabilistic tendencies and relationships between objects from the videos alone. It measures which pixels should be related, and which ones shouldn't, and so on. It is not yet trained to produce the kind of labels for videos that the human watchers produce, and is not trained on labeled data (which constitutes the "supervision" in question) at all. It is trained using only the surveillance footage (and unlabeled footage is cheaper than labeled footage, since labelers have to be paid). During this so-called *pretraining* stage, it merely analyzes raw surveillance footage, and thereby "learns" what's common and what's uncommon in the videos in general, it iteratively reconfigures itself on just raw video, to extract properties of video frames, and video frame successions, and represent them in its network, in its "wiring."

After the model has fed on tons and tons of warehouse videos, a subsequent stage of training begins, usually called *finetuning*, because it requires less data and makes smaller adjustments to the "wiring" of the previously trained (*pretrained*) model. This second stage most likely uses a different type of paradigm called *supervised learning* (a name that acquires here a strange and telling irony). In contrast to the unlabeled, raw video data from the pretraining stage, the training data used in this paradigm is labeled. In the case of the Amazon surveillance AI, this labeled data likely consists of videos that were sent to video reviewers, paired with the labels that the reviewers produced (for example, '1' if the stower in the video made a mistake, '0' if not). In the first step, the model learned the characteristics of the surveillance videos in general—now it learns to generate labels for such videos; it learns to separate out the movements desired by Amazon from those which are undesired, to identify issues, to flag people for writeups, and so on. In other words, it learns to control the stowers in the same way that the video reviewers do, by labeling surveillance videos. In this second step, the literal supervision of the warehouse workers also constitutes the "supervision" for the model training, since this second step requires not just videos,

but pairs of videos and labels. This is what the model now learns: to pair the probabilistic tendencies it extracted in the first step with the labels. That is, it learns to emulate the tasks performed by the video reviewers.

Both the warehouse workers and the video reviewers thus continually produce more of the very training data that will make the next generation of the motion tracking AI in the warehouses more accurate and efficient. The stowers' job is to stow, the video reviewers produce oversight. But in contrast to all previous warehouse operations, Amazon also manages to squeeze something additional out of both types of work: whether they want to or not, each move and each click that these workers make, simultaneously generates new data that Amazon can use in their next rounds of model training. Each new iteration of these surveillance AIs is aimed at inching that 95 percent accuracy rate ever closer to 100 percent. In this way, the video reviewers themselves are inching ever closer to being made redundant, and automated away. We are not there, yet, but AI researchers are working on making robots that can "learn from demonstration," too, so in all likelihood, Amazon's aspirations include extending the same automation to the warehouse workers, to make AIs that can move robotic arms and legs in ways that resemble the ways that stowers move today. In these ways, the work of stowers and video reviewers alike is swallowed continually by the algorithm, and made part of it. The ghost labor in Vonnegut's *Player Piano* haunted the lathe. Today's ghost labor does not merely haunt the worker and their workplace, it continually makes more ghost labor out of the labor of the living. Today's labor is tomorrow's ghost labor, the ghost itself ever growing, in an eerie echo of the growth of capital itself. What is capitalism, all told, if not past work haunting the work of the living, what is profit other than making the work of the living into ghost labor that can haunt anew.

MACHINE-MEDIATED POWER

There is something peculiar to the Amazon situation, to that particular ghost labor: it haunts the very workers from whose work it is derived. It is the defining property of capitalism that one

class, the capitalists, owns the assets that are necessary for most people to work. The assets that make up a particular company, whether they are physical, or digital, whether they are buildings, machines, data, or patents, were almost all produced by other people on some other company's payroll, by other members of the working class. Taking all workers together, as a class, we can say that the working class has made almost all the assets that are necessary for work itself. The capitalists own those assets, those things others have produced, and which are necessary for work—and they use them to make yet others produce yet more. There is a crucial asymmetry there: for the capitalists, their class position is clear and obvious—they experience themselves as individual owners of these assets, and their class as the class that owns such assets. Because of this immediate parallelism between the facts that hold at the level of class, and the facts that hold at the level of the individual, it is transparently clear to them that they are part of a class. Hence, they fraternize, sympathize, and collude with others who have the same experience. In other words, they have class consciousness, rooted in their direct experience of hiring people, and personally owning assets relevant to that. In contrast, the working class does not generally have that direct link between the individual, personal experience, and the facts that hold at the level of class. The working class may have made (almost) all of those assets, the tools and buildings, the data and the algorithms, but the individual worker does not experience this, they experience these assets only as the property of someone else, of their employer—and all the more so, the more production disappears from the social view.

But just as Mateo chillingly alluded to above, in many AI-related contexts, individual workers may now experience something directly, personally, that had formerly been only understandable if we looked at the class overall. Mateo imagined he was reviewing his own videos. The snake was eating its tail. Mateo was contributing to the building of a tool that would surveil him: now the relationship between workers and tools has become more transparent. The fact that the tool can be used to control a worker and the fact that the tool could not be built without the worker, both become clear. Usually, some group of workers produces the tools that

are then owned by another capitalist/company, in a rather dense and complex network of production, tool-use, and consumption. But here, it really is the work of those particular workers, today's work, that becomes part of tomorrow's tools, directly, without all the complexity (and obfuscation) of the global economy. In some sense, the AI, here, really is turning on those who "make it": it turns on those whose work constitutes the training data, without which no AI would be possible, and becomes a tool that manages and controls them.

How does the experience of life change for the worker, when managerial operations are transferred into machine properties, whether through a steam engine's speed setting, or through a motion-tracking AI? Of course, we cannot hope to capture or describe all aspects, but let's focus on one that is crucial to the central theme of our book—why do we fear AI? And here, this particular shift is quite revealing. We start, at first, with a relationship between two (groups of) humans, a managerial relation—that is to say, a relation of domination, where one side directs and controls the other. But after the introduction of our new technologies, be it steam power, or video classification, that relation of domination is no longer direct, but mediated: the new machine constrains the workers' activities in a way that necessitates compliance with the desires of management. In a very real sense, the workers now experience their lives as under control of the machine. That is not to say that they no longer encounter managers, nor that they do not know that someone else is making money by driving them to work faster. But in a very real sense, the machine has taken over some of the power that was previously that of a person, a manager.

Surely, one source of AI fears lies exactly here, in the fact that AIs are tools that embody the power of capital. Why do newspaper articles continue to ask whether AI will enslave humanity, whether it is some kind of evil power that bears us ill will? Why does, for example, *The Times* run headlines like "The end of humanity: will artificial intelligence free us, enslave us—or exterminate us"?¹³ Why do neoliberal pop-historians like Yuval Harari warn in *The New York Times* that AI might trap and control us "in a Matrix-like world of illusions"?¹⁴ What is it about these questions and these stories that makes them generate as many clicks as they

seem to, given that media companies run variants of the story over and over again? What makes them appealing to all those people who click on them? Surely, part of the answer is that in the real world, relations of direct domination are being replaced by machine-mediated relations of domination. Under capitalism, our tools come, in a real if incomplete sense, to dominate us. They really do insert themselves into relations of domination, and when they do, they really increase the scope of that domination—in the case of Amazon’s algorithm, to the very details of people’s physical movement, and bodily autonomy. They really are embodiments of the will-to-profit, of its “explicit verbalization,” Taylorism. No wonder that stories of AI domination, of AIs taking over the world, and trapping us in a Matrix-like nightmare, abound.

TAYLORISM CRUSHES IT: GENERATIVE AI AND DESKILLING THE LONG TAIL

In the summer of 2024, Apple aired a new video ad for its latest tablet iteration.¹⁵ The ad begins by showing us a collection of tools, whose use clearly requires skill and knowledge, arranged into a sleek pyramid. Most of the pieces are art-related. A trumpet crowns the arrangement. A metronome ticks implacably and ominously, perhaps as if saying “Your time is running out.” There are other instruments, like a piano and a drum set, alongside studio monitors, and a mixing table. Tools of the visual arts are equally prominently displayed: various paints, a poseable artist’s mannequin and easels, a film clapper and cameras. An architectural diorama, and notebooks also have their place, as does a blackboard with some nonsensical (?) math. Alas, the arrangement sits atop the platform of a hydraulic press, and gets slowly, painstakingly crushed. Paint gushes, piano keys shear and splinter, glass shards of camera lenses spray out explosively, the trumpet gets bent and contorts into an unrecognizable knot. When the press lifts up again, all that remains is a tablet. All the tools, all the skills, art and love they represent, have been symbolically—violently—transferred into the iPad.

Apple's ad triggered a wave of backlash. It had been all too real. The aim of transferring knowledge and skills into tools was laid perhaps a little too bare. The ad, called "Crush!" (even the name said the silent thing out loud), inadvertently gave the game away: in transferring all the skills represented by all those tools and materials into the tablet, human agency and the human meaning derived from honing and using those skills, got bulldozed, devalued, and disappeared. What used to be not just any skills, but skills deeply representative of the possibility of taking pride and pleasure in one's work, was flattened into yet another commodity—"an absolute powerhouse for AI" available on apple.com, starting at \$999.

The advertisers may have been oblivious to it, but to plenty of people one thing was obvious: the Apple ad represented a devaluing of knowledge and skills. Taylorism strives to centralize knowledge in a managerial bureaucracy, in order to separate out tasks, and deskill them. The Apple ad symbolizes a new and blatant form of techno-Taylorist deskilling—the transfer of knowledge and skills not into a managerial bureaucracy, but directly into a tool, into a machine. In this case, the iPad took center stage—but everyone understood that this ad was about that one flashy technology that is most emblematic of artificial intelligence today: generative AI.

Think about generative AIs for art and design that generate images from text input, like Dall-E or Midjourney (trained, as so often, on the private property of millions of artists, and on global as well as local commons). Think about AIs for writing. Think about AIs that are supposed to find and summarize legal cases. Think about AIs aimed at writing computer code, like Microsoft's Copilot. Think about education AIs that are supposed to aid or replace tutors. The list of domains that generative AIs are being developed for can seem endless. Any task whose aim is the production of digital objects—from text to audio, from images to movement instructions for a robotic arm, is apparently up for grabs by generative AI.

One can and should harbor doubts about how successful models based on the current crop of machine learning techniques will ultimately be—but there is little reason to doubt what the

aim is: to embed knowledge in machines—knowledge that previously belonged to the people working in these domains. The nature of the human knowledge that is targeted varies greatly, within and across domains. It might be abstract or concrete, it may be declarative or procedural. But regardless of its nature in these or other regards—from an economic perspective, it is usually knowledge that counts as potentially marketable skills, knowledge that is constitutive of the primary asset that the workers in these domains have to sell: their ability to perform particular work. In other words, just as Amazon’s warehouse AI embedded a surveillance drive of management in a machine, so generative AI embeds knowledge of tasks (and the drive to centralize it) in a machine. Both the surveillance AI and the various generative AI approaches realize the same drive towards lower wages that already brought the steam engine into the cities—the need to make workers replaceable by others, to lower wages by increasing labor supply. The method of generative AI is technological Taylorism. The point is—to deskill.

MAKING GOOD JOBS BAD

What kinds of effects will this deskilled, “AI-powered” work have? The question is probably worth its own book, but one thing that seems obvious, is that it will make things cheaper—both in terms of price, and in terms of quality—and that this will further exacerbate pre-existing inequalities in many of the domains targeted by generative AI. Most work that can be deskilled by generative AI will be increasingly gigified, while a small section of specialists will produce AI-free and “artisanally” for the upper crust. So-called self-driving cars already operate on a mix of automation and gig-work. The “autonomous” vehicles of General Motors subsidiary Cruise, for instance, are remotely assisted some 2–4 percent of the time, or every 2.5–5 miles, with their human “autonomous vehicle operators” continually monitoring many cars simultaneously.¹⁶ The rich, of course, have their driving produced artisanally already—by chauffeurs. AI tutors will be running on language models, but just as with Amazon’s warehouse surveillance and autonomous cars, AI tutoring will also be supported by a hidden underclass of deskilled

gig-workers that monitor the chat windows of dozens of students at the same time. Meanwhile, the wealthy will continue to hire Ivy League PhD students to look after their children.

The legal domain, just like education, is already starkly divided for the poor and the rich: a wealthy defendant may well have a whole team of people working full time on their case, while poor defendants will be lumped in with a dozen others on a public defender's docket. AI legal assistants and AI legal search engines will likely lower the wages for paralegals (who are being deskilled). "Ambulance chasers" and "sharks" will increase their dominance of the low-end market, cutting costs due to AI, while also undermining quality. Competitive pressures mean that honest and dishonest lawyers alike will have to take on the absolute maximum number of cases that can be done in a certain amount of time. Those lawyers who genuinely care about serving the underserved will suffer from the increasing pressures of this competition in their market: AI-assisted competition will drive up the numbers of cases these lawyers have to take on, and drive down the quality that they can afford to provide. Meanwhile, wealthy clients and corporations will be largely unaffected, since they will continue to be served by law firms whose services are way out of reach for almost everyone else.

In the realm of coding, big tech companies will continue to pay high wages to those who produce code for high stakes situations, where even small optimizations can be worth a lot of money. But an increasing number of programmers will be subjected to competition from companies that provide minimal training (say, a four-week course in using generative AI for programming), and that treat their workers as replaceable and disposable, gigifying and deskilling ever more coding work.

In sum, in many of the domains where generative AI may play a major role, we think the effects of AI automation and deskilling will likely be an acceleration, intensification, and spreading of tendencies that we can already observe today, such as gigification: it will make low-quality work somewhat worse, but much cheaper, thus creating downward pressure on wages and quality alike, while putting high-quality work increasingly out of reach. Out of reach, both for all those workers who would much rather do such work,

and for those consumers who cannot afford to buy anything but things and services produced in AI-assisted contexts.

Generative AI will bring this kind of crowding out of the mid-level quality products, and the (often good) jobs associated with them to new parts of the economy. At the same time, the pattern is a more general one that has been quite typical of capitalist development for decades now, and another aspect to the deskilling drive. From shoes to tools to pots and pans—the medium-quality market has been crowded out by a lower-quality level that is often much cheaper (and most of the time simultaneously less durable), while the higher level quality goods are increasingly out of reach for most people. Perhaps the most obvious example is furniture—ready-to-assemble furniture from IKEA and its copycats are ubiquitous, but furniture that could last for more than one move is increasingly limited to a high-end market. The high-end market may be largely unaffected, but the mid-level quality often cannot compete with the lower end. This is especially true in a country like the US where the majority of people live paycheck-to-paycheck and find saving almost impossible. Often, cheaper goods have to be replaced significantly more frequently, perhaps even to the point where the cheaper goods end up costing more money in the long run. It's expensive to be poor.

Clearly, all of this will further stratify the working class. It will create more separation between “good” and “bad” jobs and increasingly differentiate pay levels. It will make the experience of being a worker under capitalism ever more different according to which stratum of the working class one is in. The stratification of tomorrow grows directly out of the stratification of today, and deskilling is, in fact, a major reason for working class stratification as it exists now. Already, in many professions, we find a spectrum from highly deskilled work aimed at producing lower cost goods, to workers with deep knowledge of the whole labor process, who produce high-end, or luxury commodities and who have higher incomes. Consider, say, a watchmaker. While most of us use watches mass produced in industrial settings (or simply our phones), prices for “artisan-made” luxury watches can easily outpace the lifetime earnings of most workers. Differentiation-by-deskilling.

What the spectrum from deskilled to “artisanal” work looks like, both from the producer’s and the consumer’s perspective, is of course quite specific to the particular object. We could contrast watches, for instance, with restaurant meals. In restaurants, deskilling is often achieved simply through the organization of work, by management. That is, tasks are split up, and assigned to people of different wage levels—say, bussers, prep cooks, line cooks, sous chefs etc. Again, it’s important to stress here that “deskilled” is a technical term that has little to do with whether a worker has skill, and far more to do with how easy it is to replace them in a particular position: any prep cook is highly trained and skilled in a number of domains, and most can easily out-chop anyone without that training; in fact, these skills are a job requirement, whether the position is deskilled through organization/management or not. But the way that the labor process is divided is aimed at deskilling the prep cook, and concentrating knowledge of the whole of the labor process with management. In the case of a restaurant, management might simply be the chef manager, who sets the restaurant’s menu, is relevant to the restaurant’s brand, etc. Of course, the chef manager may well be the actual capitalist of their own small capital, if we are talking about a single restaurant. In the fast food industry, i.e., in the most gified part of the restaurant industry with the most mass-produced (and often cheapest) food, organizational deskilling is accompanied by technological deskilling. Burger patties and french fries, for instance, may be prepared in fully industrial settings, and arrive at the restaurants in frozen form. Since this allows for intense standardization and maximum task splitting, the work performed by those restaurant workers, too, is maximally deskilled—that is, their performance and their movements can be tracked according to detailed standards, and management can often treat people not as humans, but as replaceable inputs in their production process.

Even in the general public discourse, the derogatory “flipping burgers” is often used to denote bad jobs that have been subjected to deskilling to a very high degree—but of course that was never in the hands of the individuals who work there. Deskilling, continually chased after by capital, produces a simultaneous need for a labor pool deprived of other opportunities. Fast food workers,

of course, have correctly identified these facts, and are joining forces. They have been courageously fighting back, for instance through forming the Fight for \$15 movement that demands a right to a union and a higher minimum wage.

The transfer of knowledge into tools that is involved in deskilling also produces a second source of working class stratification: whenever knowledge is embedded into tools, an associated kind of specialist emerges, people who do have access to the relevant knowledge, who can further develop the tools, who can service and customize them. In the case of the steam engine, this included, for example, the machinists who could make the relevant parts. Of course, just as with the pyramid of managers, the expanding specialist labor may subsequently itself be subject to deskilling, as happened with machinists. But broadly speaking, deskilling tends to be accompanied by the creation of its own techno-bureaucracy, responsible both for the tools, and for concentrating knowledge relevant to them. Sometimes, the members of this new bureaucracy are first drawn from the workers themselves, sometimes they come from outside.* Regardless, they are responsible for putting into the service of capital both the knowledge embedded in the tools, and the managerial knowledge of the tasks (and their splitting up) that the tools require. In other words, they will form a specialized group whose job is in an ambiguous field between engineering and management, as evident for example in the many degrees in fields like “engineering management,” “business engineering,” “technology management,” “systems engineering” (a term that can, perhaps not incidentally, be traced back

* Indeed, Frederick Winslow Taylor himself became a manager, and then the founder of scientific management on such a path. While from a wealthy family (his father was a Princeton-educated lawyer), Taylor decided to forgo his Harvard admission, and become a machinist instead. After an apprenticeship, he started out as a workman at Midvale Steel Works. Then, however, he became the gang boss of his former coworkers, who soon called him a tyrant. From the moment he took on that position, he continued to develop his basic ideas: that any control that workers have over the processes of work stand in the way of productivity, of capital pressing the most labor out of any person hired, and that hence, all such knowledge must be concentrated in management.

to Bell Labs, the birthplace of the transistor), and many more like them. Any company will happily pay higher wages for these “upskilled” workers, as long as it makes up for the lowering in wages, and the potential increase in speed, control, productivity, and so on elsewhere. Perhaps a company could either pay ten workers 50k each, without deskilling them, or it could pay one worker (turned techno-bureaucrat) double, and nine deskilled workers half. This is the real life version of the industry fable of the so-called “10x engineer,” a mythical engineer that allegedly adds 10 times the value of a normal one to a company, and the real mechanism behind it: the value that is “added” is literally the wages that the other nine workers are no longer paid, and which thus remain on the credit side of the company’s ledger. Whenever new technology enables such an equation to be drawn up, capital will happily produce highly paid positions—provided, it saves enough money elsewhere.

As we live in times where the promise of social mobility through education is increasingly turning stale, with many people, particularly in the US, buried under suffocating piles of student loans, perhaps it’s worth reconsidering whether that promise wasn’t always based on the fact that specialists are often the flipside of a deskilling process for others. That is, perhaps it was always impossible for that path to be plausibly available for everyone—after all, most people have to be on the deskilled side for the equation to work out in capital’s favor.

MAKING TOOLMAKING TOOLS

What does transfer of knowledge look like on the side of the machines? At a very abstract level, this is simply the transition from special purpose to (more) general purpose machines, or the ever increasing ability of machines to be programmed, and to process more and more information. And indeed, that would be an excellent way of re-telling the story of the transistor. Although written as early as 1974, we can draw, again, on Harry Braverman’s book—perhaps sometimes historical processes are easier to understand early on, than when they have developed all kinds of complexities. In discussing automation in the metal industry,

Braverman (himself a former metal worker) suggests that automation first targets objects that are produced on a very large scale: it tends to be worth investing into machines for making things like screws or standard car parts. But those screw machines themselves contain many specialized parts, each made only a few times. Since we need far fewer screw machines than screws, screw machine parts in turn will be made in much smaller batches (some perhaps will even be unique). As it turns out, far more objects are produced in small batches (like screw machine parts), than in large batches (like screws). And because they are produced only in smaller quantities, it is harder to automate their production cost-effectively.

Imagine a graph of all objects produced in a year, ordered by how many times they are produced. Picture the graph with count of individual objects produced on the Y-axis, and the individual objects on the X-axis. Let's say the objects are listed in increasing order of production count. Our graph would start with a very long list of objects that were produced only once, then a long list of objects that were produced twice, etc., until it ends with the object produced most often. A distribution like that is said to have a *long tail* of many, many objects produced in small numbers. Automation will tend to target the right side of the graph first, aimed at those objects that are produced in huge numbers, and it will involve special purpose machines, like a screw machine, designed for the production of that one object, in massive quantities.

What about the long tail, all the objects that are produced only once or a few times? This is often a place for knowledgeable specialists, people who have the skills to use a variety of tools to produce highly specific results. In the case of our screw machine parts, it might have been machinists who knew how to precisely cut metal into the complex shapes needed. But the necessity of that knowledge increased the machinists' bargaining power, and hence tended to produce higher wages for them. We know, by now, what shape capital's solution to such a "problem" takes: transfer knowledge to the machine! In this case, such a transfer of knowledge was originally known as *numerical control*, i.e., machines that could process a certain amount of numerical information.

On the one hand, machines were equipped with ever more monitoring devices and sensors that could provide information about the current state of their task. On the other hand, machines were increasingly programmable, originally with punch cards—the same kind of technology that allows for player pianos to be programmed to play different songs (as in Vonnegut’s novel)—then increasingly with transistor-based chips. Numerical control machines turned into computer numerical control machines, or CNC machines, which could cut increasingly complex shapes. They could take over more and more calculations (say, the path along which a cutting tool must move to create a particular curvature, radius, etc.) that had previously been part of the machinist’s job. Because they were programmable and increasingly information processing, they could fulfill an increasingly larger set of functions, and hence make an increasing variety of objects. Today, even YouTube is full of videos of CNC machines carving wood or metal blocks into fascinating and unique objects, from screws that turn both ways to spectacular marble runs. Unlike our screw machine, then, these machines can make a great variety of pieces. Or, to put it another way: as we go down the long tail, we move from objects created in large numbers by special purpose machines to objects created in small numbers by (more) general purpose machines.

Automation by AI is very much a continuation of this trend. Generative AI is aimed at automating the production of many things at the very end of the tail, from newspaper articles to logo designs—that is, things that are really only produced once. Take even the term “general” in “artificial general intelligence.” Clearly, if AGI has a non-occult meaning (i.e., if it still denotes a tool rather than a myth) the “general” in it is the one from “general purpose.” The development of such a fully general purpose machine might be considered the holy grail of AI research and development. Even “ChatGPT” might be a pun on this: while GPT stands here for *generative pretrained transformer*, a language model architecture, in economics GPT is often short for *general purpose technology*, technologies with extremely wide applications like the steam engine, electricity, or the transistor. And while that may or may not have been a sneaky reference to that envisioned

general purpose character, sometimes the thing is said quite explicitly. Take the definition from the charter of OpenAI: “artificial general intelligence (AGI)—by which we mean highly autonomous systems that outperform humans at most economically valuable work.”¹⁷ Clearly, they are talking about a general purpose machine, one that could (if it existed) replace *homo sapiens* the toolmaker with itself, the toolmaking tool.

MACHINES DESTINED TO BE DISPOSED OF

A world dominated by such hypothetical tools would seem to have little room for humans to act purposefully on the world, or derive any meaning from life at all. As OpenAI CEO Sam Altman once said: “AI will probably most likely lead to the end of the world, but in the meantime, there’ll be great companies.”¹⁸ If the aim seems absurd, we must remember that it represents the logical endpoint of one of capitalism’s tendencies, the drive to deskill, to transfer knowledge from workers to machines, from humans to capital. Speculations about AGI are, in this sense, speculations about the endpoint of the drive to deskill: ultimately targeting *all* human knowledge. That is not to say that this is a plausible outcome; there are many countervailing forces (not least the fact that we continually generate new knowledge and new tasks). Indeed, capitalism’s relationship to knowledge is rather paradoxical: on the one hand, there is a tendency to universalize certain kinds of knowledge, to make sure everyone is in possession of it—since that, too, is a form of deskilling. That is, if everyone knows a thing, then no group of workers can gain special bargaining power from possessing that knowledge. We can take reading and writing as an example, and we could look at many lobbying efforts and political arguments over what should be included in school curricula through that lens. Even the investments that many big tech companies have been making into their own kind of schools, so-called coding academies, can be understood in the light of these universalizing pressures: make everyone into a programmer, and programmers can no longer demand higher-than-average wages.

On the other hand, capitalism produces precisely the opposite relation to knowledge, one that is decidedly not that of an

open and democratic society. We might say that capitalism's other relation to knowledge resembles nothing so much as an intelligence agency. That is, it is incessantly involved in gathering and producing knowledge, while also putting all that knowledge on a "need to know" level of security, concentrating it, and sharing it only when its aim (profit) cannot be otherwise achieved. This is the movement that continually transfers knowledge from workers into tools, from people into capital. If the "general" in AGI is the one from "general purpose," then this, surely, is the actual meaning of the "intelligence" in A(G)I: the kind of intelligence you gather, not the kind that you possess.

Braverman described this second aspect, the use of science in ways entirely antithetical to science itself: "The more science is incorporated into the labor process, the less the worker understands of the process; the more sophisticated an intellectual product the machine becomes, the less control and comprehension of the machine the worker has." These two ways of treating knowledge can seem utterly contradictory—only if we conceive of them both as forms of deskilling, of undermining the bargaining power of workers, can we make sense of this apparent paradox.

Let's take another look at that definition of AGI—"highly autonomous systems that outperform humans at most economically valuable work." That "outperform" is a rather telling word, isn't it? Under no organization of society, other than capitalism, are the tools somehow conceived of as being in competition with the tool-users. The philosopher Shannon Vallor has discussed this idea of competition, and the way it impoverishes our sense of self, our idea of meaning, and even what it means to be human: "those invested in the maximally efficient extraction of productive outputs from human bodies have been trying to get us to view ourselves—and more importantly one another—as flawed, inefficient, fungible machines destined to be disposed of as soon as our output rate slips below an expected peak or the moment a more productive machine can be found to step in."¹⁹ She's certainly right. We might even go one step further—it is not simply that those invested in the maximally efficient extraction of productive outputs (those that we call capitalists or the representatives of capital in this book) try to get us to view ourselves this way.

It is also that this reduction of our humanity is inherent to the semi-voluntary mode of “cooperation” that is wage labor; to capital, we are always a means to an end (profit), and never partners in a project that would involve its own meaning (regardless of what HR departments may tell us). In chasing profits, in seeking to maximize its measure of success, capitalism must seek to press the most productivity out of tools and people alike. And what is capital’s drive to deskilling, if not the drive to make us ever more fungible (interchangeable and substitutable)?

Of course, the fungibility that deskilling is aimed at, is about making workers replaceable by other workers, while automation is ultimately aimed at substituting tools for people. In that sense, under capitalism, the competition between worker and tool that the word “outperform” implies is real. When a task is partly automated, when productivity is increased, the people who used to do all that now superfluous work do not generally rejoice that they have won free time to spend on something else. Instead, the people themselves are made superfluous to the enterprise they were formerly engaged in. That is what, in Vallor’s words, turns people into “machines destined to be disposed of.” Some, maybe even all of the replaced people get fired, and the savings in time become the source for extra profits for capitalists, and extra distress for the workers.

One of the truly strange properties of capitalism, isn’t it? That it continually makes people superfluous, rather than merely their work, setting people free not from boring toil, but from their ability to make a living. In a society that would deserve the label “free,” perhaps automation would be aimed not at making people superfluous, but at eliminating the aspects of work that are onerous, that feel like drudgery. And all the scientific and engineering efforts that go into automation could be aimed not at deskilling, not as something set against the workers, but aimed at increasing the creative, expressive, social, scientific, and otherwise fulfilling parts of work. It could increase human agency over the labor process, rather than concentrating it within managerial structures and machines. Alas, capitalism is no such free society, and its drives push in a wholly different direction.

THE WORLD IS NOT TOO COMPLICATED

We have now explored AI as a continuation of many previous technological, economic, and political tendencies. Now we've arrived at a point of payoff, so to speak. Implicit in our perspective were kernels for explaining a number of other phenomena that we've been thinking about throughout this book: why did the internal functioning of our tools increasingly vanish from the social view? Why have bureaucratic-managerial structures ballooned wherever we look? Why is the working class so stratified? Why is it that the more humanity's technological and scientific powers increase, the less we feel like we can actually control those powers? And perhaps we can even shed some new light on why it is so hard to imagine an alternative to capitalism.

The story of the transistor illustrated the changing nature of our tools: technological developments have increasingly severed our relationship to our tools' internal functioning. We suggested that we experience this as an increase in complexity. But what is that increasing and mysterious "complexity" other than knowledge that is purposefully embedded in the machine or the managerial apparatuses, in order to make it unnecessary for workers and users, for us, to understand it. After all, for every complex aspect of our technology, there is *someone* who understands, who does know. Complexity is decidedly not the same as genuine incomprehensibility. That is not to say that the scale and differentiation of new scientific and technological knowledge have not played a part. But under capitalism, new knowledge, and in particular knowledge of tools and labor processes is continually and systematically put on a need-to-know basis, and often sealed off, as an effect of deskilling.

Knowledge gets concentrated among managers or technobureaucrats, so as to deskill, to cheapen the labor of others. The unequal distribution of knowledge in society is an effect of deskilling—and, insofar as that unequally distributed knowledge is also continually used in new technological and scientific developments, it also produces the affordances for new rounds of deskilling. In other words, under the pressures of capitalism, technological developments and the sealing off of knowledge are in a self-reinforcing feedback loop. And since all that knowledge is systematically put

on a need-to-know basis, monopolized by some, it cannot appear as anything other than complexity to everyone else—as knowledge possessed only by someone (or even something) else.

The vanishing of our tools' comprehensibility is not simply an accidental byproduct of technological development. The closing off of our tools from view serves a particular purpose—to increase the power of one class, the owners of those tools, over the other one, those who work with them. These economic pressures and developments end up affecting all kinds of tools, whether they are capital or consumer goods. From computers to DNA testing services, many things do, after all, transition from one domain to the other, as their prices fall, and new uses are invented. If the backs of our smartphones are literally and figuratively glued shut, then this is not simply an inevitable side effect of technological development, but about locking knowledge behind closed doors, and sealed backs, about putting knowledge on a need-to-know basis. In other words, it is the fact that knowledge is power—the power to bargain for higher wages, and better working conditions, or the power to decrease wages, and increase capital's profit and control over work—that made our tools into what they are: objects whose complexity enables us to be ever more productive, but that also continually shift power from workers to capital.

The vanishing of our tool's internal functioning went hand in hand with a rise of innumerable and ever larger bureaucracies, even as we lived in times dominated by a neoliberal self-justification of capitalism that perpetually claims to be opposed to bureaucracies. Such justifications even claim that capitalism and bureaucracies are like oil and water, almost akin to logical opposites. Now, we know that the freest of capitalist markets isn't inimical to bureaucracies that hoard knowledge in order to create control. Quite the contrary—capitalism incessantly and by necessity produces such bureaucracies. Competition drives corporations to continually develop new technologies, in order to lower production costs per piece. This can be achieved not only through increased productivity, but also through deskilling—to the capitalist, they are simply cost reductions, reductions in input variables, indistinguishable from the only thing that ultimately matters: prices and profits. And thus, wherever new technology makes increases

in productivity economically feasible, that's what it will do—and wherever deskilling is economically feasible, *that's* what it will do.

Deskilling, however, requires that knowledge be put on a need-to-know basis, transferred away from workers and embedded in managerial and specialist bureaucracies. At first, this may only create managerial bureaucracies. But soon, even engineers that started out as purely technical specialists are more and more involved in the planning of production processes, in gathering knowledge to exercise control. Wherever oversight is transformed into a technological property of tools (from the steam engine to the Amazon AI), engineers become techno-managers. Nowhere are engineering departments concerned solely with productivity, they are concerned with costs per piece, and hence they turn into techno-bureaucracies. Management and technical specialists start forming mixed bureaucracies: technology and management, tools and control are ever more deeply intertwined. And whenever those bureaucratic structures get big enough, the same principles of deskilling are applied internally as well. A process of recursive deskilling and bureaucracy creation unfolds, and is unfolding every day. The managerial pyramid turns into a professional-managerial pyramid. It grows incessantly, as long as it can keep overall wages down and make sure that the ever increasing productivity is turned into profit—and never into social wealth, into more interesting and engaging workplaces, or into simply more free time.

We began with a very abstract description of what class is: to be a member of the working class is to live off wages, to sell one's ability to work, so that someone else can put it to use for their profit. To be a member of the capitalist class is to live off the profits generated by capital, by assets (often the kind of assets that require someone else's work), rather than by selling one's ability to work. But of course, the real, lived experience of class is much more varied than that, the class of people who depend on wages is not internally uniform, but extremely stratified. In fact, among people who work for a wage in the US, the majority would likely consider themselves "middle class," and not "working class." We can now connect those two kinds of descriptions—stratification is inherent to the professional-managerial pyramid, which concentrates knowledge and control over work processes,

at every level. It's the recursive nature of the drive to deskill that gives rise to stratification, as it applies wherever profits and control can be increased by deskilling many and upskilling a few. In divvying up labor processes, capitalism increasingly divvies up the working class itself.

The more divided the working class is, the harder it is for them to find a common interest and common ground. This is obvious at the level of wages (where division undermines collective bargaining power, by separating out different interests). But it also applies at the level of politics, which is dominated by shifting alliances that sections of capital make with strata of the working class. Of course, we have only touched upon many of these developments, and viewing the working class as stratified by their access to economically valuable knowledge has many potentially useful applications that we won't be able to get to in this book. Consider, for instance, the right-wing attempts to channel anger among some working-class people away from collective conflict with their bosses, and into anti-intellectualism. There are ongoing attempts from the right to declare a barista with a college degree and \$100k of student debt the "elite," while the inheritor of a used car dealership, a small capitalist that makes seven figures in profit per year, is declared "working class."

Earlier in this book, we had asked why humanity, having gained the awesome power to change the climate of our whole planet, simultaneously seems to have lost the power *not* to change it. Or more broadly, why is it that the more scientific and technological power we gain, the less able we seem to control those powers? In that chapter, we talked about the way that capitalism imbues our tools, as capital, with a kind of motive force of their own, the drive for profit. There, we said that this drive for profit is a structural pressure, an organizing principle of our society—it could not exist without a capitalist class, without managerial executors, but capital's drive to profit does not depend on any particular person's desires (it even shapes the desires of people to match its own requirements). Or in other words: capitalism is a system that dominates everyone inside of it, even the capitalist class is subject to its pressures. And when tools are capital, they are things that exert that pressure, the pressure to use them to

make a profit, wherever and whenever possible—even if it means trashing the climate.

We can now add a second layer to that account: from the perspective of the working class, tools really do represent not the power to do something, but the power that others (the capitalist class, and their professional-managerial representatives) have over them. We quoted Harry Braverman's paradox earlier: "The more science is incorporated into the labor process, the less the worker understands of the process." This is the mystery solved—why does an increase in our scientific understanding limit, rather than enhance, our ability to act on the world in accordance with our will? Why have we gained awesome powers, but lost the power to not use them? Because the science and the tools that grant humanity these powers, are simultaneously weapons wielded by one class of people, the class of capitalists, in order to control everyone else in the name of profit. The manifold applications of science, the tools that capitalism continually develops are also Taylor(ism)-made to control and cheapen people's labor. Science is used to hide, privatize, and monopolize science. The more knowledge we create, the more effective the capitalist class becomes at putting much of it on a need-to-know basis. The more sophisticated the tool gets, the more bureaucracies pop up to control it for capital, and the more its internal workings are hidden from the social view. The more science is embedded in our tools and labor processes, the less we can control them. They, instead, control us, so that capital can make its most profitable use of the one thing that it buys from us—our ability to work. In other words, the 'humanity' that has gained the power to change the planet and has lost the power not to, that's the capitalist class. The capitalist class has gained power, but it really is power that they cannot not use (at least not if they want to remain capitalists). It's a power wielded against the 'nonhumanity,' the working class, the rest of us—we have never lost it, we have simply not yet taken it away from those who want to use it to make a profit, no matter what.

Finally, why did so many of us end up believing Thatcher's dictum that "there is no alternative"? Where did the old utopias of a deeply democratic organization of society and of work disappear to? Why do markets, wage labor, and profits seem like

the only way to organize society, even though it makes so many lives miserable, and is threatening to make our planet uninhabitable for humans? Why can we no longer imagine that we could, simply put, make do without bosses, without managers, without capitalists? Why could neoliberalism—in the words of Naomi Klein—“convince us that we are not just incapable of self-preservation but fundamentally not worth saving”?²⁰ Why could it convince us that only the market, only capital can know how to organize the world? Because capital has indeed monopolized the knowledge of how the world is organized today, concealing knowledge wherever it can, revealing only what it must. By embedding knowledge in tools, capitalism privatizes not just the instruments that we use in making the many things that we need to live, from food, to housing, to art, to care. If the tools’ internal structure is hidden from view, privatized, its workings put on a need-to-know basis, how can we, the class that makes all the tools, imagine ourselves as making do without capital? As the internals of our tools vanished, so did the fact that all the things we consume have been produced by us, by the working class. We live in a world where goods seem to simply appear in the store. Many of them do so in ways that make it not only unnecessary to understand how they were made, how they got there, but difficult, or even impossible. This, too, is an effect of deskilling, of privatized knowledge—the sense that objects no longer have a history of their production *by us*.

And yet, as we understand these issues, as we unveil the relation between economics and power, between tools and politics, we can also discover new possible paths. As workers, we must struggle collectively not only for wages, but for our ability to directly control and comprehend the labor processes that we are engaged in. Those are the sources from which we can develop our collective strength, the strength we need even in the struggle over wages. They are also the locus at which economic struggles become clear as political struggles—not just struggles over how we distribute the pie, but how we decide on which pies to make, and whether the world should really be an oven.

Wherever we form or join unions, we should strive to elevate the collective control over knowledge, the democratization of it in

the workplace as one of our main goals. It is up to all of us to take on this task, to develop better ways of sharing, distributing knowledge, of de-privatizing it. We need to make knowledge into tools of our collective agency, rather than private tools for deskilling, stratification, and capitalist control. We need democratic control over and collective understanding of our own labor processes.

The fact that the world appears as “too complicated” is a real veil that is at the heart of the disappearance of classical utopias. But we do not need to stand behind it. Capital’s power to monopolize knowledge is not based on some mysterious power that only it can wield, it comes from organization and is based in bureaucracies. If society decided to (if the working class decided to), we could abolish corporations, take them over, and make them into institutions that are cooperatively run by the workers, run so as to free knowledge, not to hide it, to give people knowledge and control over their own work, rather than centralizing knowledge and control in capital. This is a question of imagination, and of collective will—a question for the political animal, not for the toolmaker. We can change the world, but we can only do so together. We need to organize ourselves, and gather the knowledge with which we can build a better world.